Questions on Artificial Intelligence

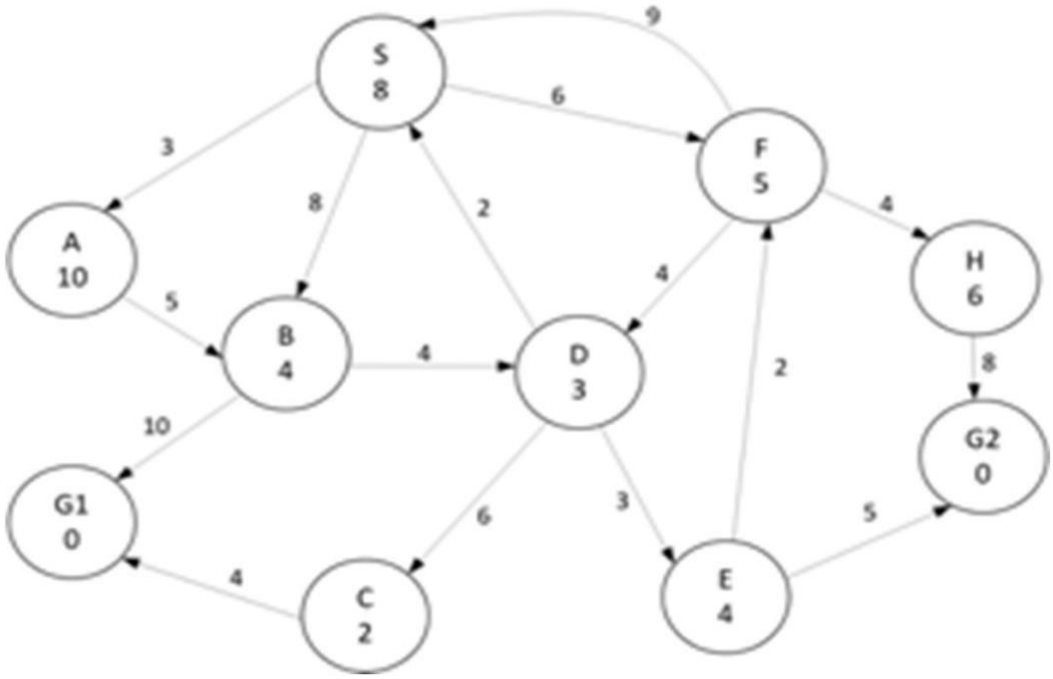
Q1. The A* algorithm is optimal when,

a. It always finds the solution with the lowest total cost if the heuristic‘h’ is admissible.
b. Always finds the solution with the highest total cost if the heuristic‘h’ is admissible.
c. Finds the solution with the lowest total cost if the heuristic ‘h’ is notadmissible.
d. It always finds the solution with the highest total cost if the heuristic‘h’ is not admissible.

Q2. An A* algorithm is a heuristic search technique which

a. is like a depth-first search where most promising child is selected forexpansion
b. generates all successor nodes and computes an estimate of distance (cost) from start node to a goal node through each of the successors. It then chooses the successor with shortest cost.
c. saves all path lengths (cost) from start node to all generated nodes and chooses shortest path for further expansion.
d. None of the above

Q3. Consider the search space depicted in the Figure below. S is the initial state. G1 and G2 are two states that satisfy the goal test. The costof traversing from one state to another is depicted by the numerical value close to the edge connecting the two states. The estimated cost tothe goal is reported inside the states. Use alphabetical order of nodes tobreak ties. Which goal state is reached if you perform A* (graph)search? What is the largest value that the heuristic function can take fornode A while still being admissible?



a. G1 and 16

b. G1 and 15

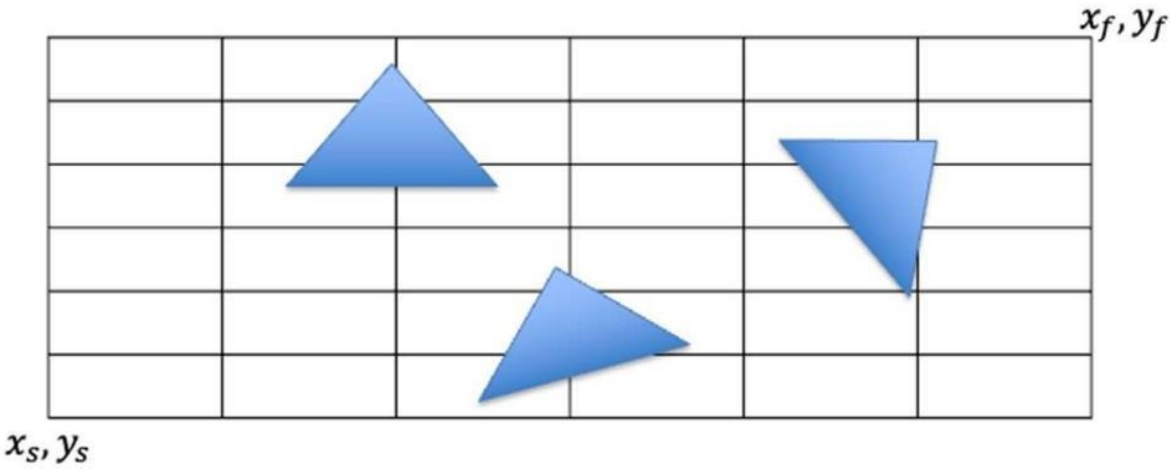
c. G2 and 16

d. G1 and 15

Q4. Consider the grid world shown in the figure below. An agent is planning to move from the starting location (x_s, y_s) to the final location(x_f, y_f). The obstacles along the path are triangular in form. Consider the following heuristic functions to conduct A* search.

- h_c assumes the obstacles are the smallest circles circumscribing the triangles.
- h_r assumes the obstacles are smallest rectangles circumscribing the triangles.
- h_c' assumes the obstacles are largest circles inscribed in the triangles.

- h_r' assumes the obstacles are largest rectangles inscribed the triangles.Which of the following statement(s) is (are) true?



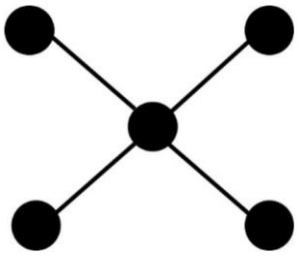
A. h_c is an admissible heuristic

B. h_r is an admissible heuristic

C. h_c' is an admissible heuristic

D. h_r' is an admissible heuristic

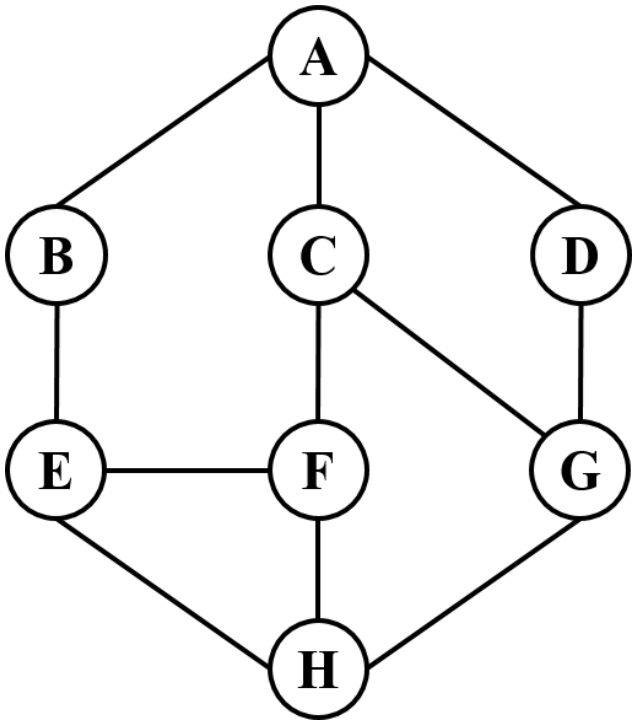
Q5. Consider the following undirected graph on 5 nodes



Assume you are performing breadth first search on this graph using a queue data structure. How many unique breadth first orderings are possible on this graph?

A. 9
B. 24
C. 48
D. 120

Q6.



A→H – 45
B→H – 40
C→H – 18
D→H – 35
E→H – 25
F→H – 12
G→H – 10
H→H – 0

Given Best First Search with heuristic value. Starting point is A. andGoal state is H. Find the Best solution path.

A. ACFH
B. ABEH
C. ACGH
D. ADGH

Questions on Machine learning

Q1. What is the primary idea behind ensemble learning?

A. Reducing model complexity

B. Combining the predictions of multiple models

C. Increasing the training data size

D. Minimizing overfitting

Q2. Which ensemble method trains multiple models sequentially, witheach model correcting the errors of the previous one?

A. Boosting B. Bagging

C. Stacking D. Random Forest

Q3. In Random Forest, what is the primary technique used forcombining the predictions of individual trees?

A. Weighted averaging B. Stacking

C. Majority voting D. Gradient boosting

Q4. What is bagging in the context of ensemble learning?

A. Sequentially training models to correct errors

B. Training multiple models on different subsets of the data

C. Combining the predictions of models with weighted averages

D. Training models with increasing complexity

Q5. Which of the following is an advantage of ensemble learning?

A. Increased risk of overfitting

B. Higher computation time

C. Improved generalization performance

D. Limited diversity of models

Q6. Suppose you have a dataset with 1000 samples. If you create 10 subsets (bags) with replacement and train a model on each subset, howmany samples, on average, would each model see during training?

A. 1000 B. 100
C. 500 D. 10

Q7. In boosting, if a model's prediction error is higher than expected, how are the weights of misclassified samples typically adjusted for thenext model?

A. Increased B. Decreased
C. Unchanged D. Doubled

Q8. What is bagging primarily designed to reduce in machine learningmodels?

A. Bias
B. Variance
C. Underfitting
D. Overfitting

Q9. If a dataset has outliers, how can bagging help in terms of modelperformance?

A. It cannot help with outliers
B. It may amplify the impact of outliers
C. It tends to ignore outliers during training
D. It may reduce the impact of outliers

Ans:

1. B 2. A 3. C 4. B 5. C 6. D 7. A 8. B 9. D

Questions on Machine learning

Q1. Which of the following statements best describes supervised learning?

(a) The model learns from unlabeled data.
(b) The model learns from labeled data and makes predictions on new,unseen data.
(c) The model learns without any form of guidance.
(d) The model learns from reinforcement signals.

Q2. In a supervised learning problem, what is the role of the target variable?

(a) It is used as a feature to train the model.

(b) It is used for unsupervised learning.

(c) It is an irrelevant factor that should be ignored.

(d) It is the output variable that the model learns to predict.

Q3. What is the main goal of a regression task in supervised learning?

(a) To classify data into different categories.

(b) To cluster similar data points.

(c) To minimize the error between predicted and actual continuous values.

(d) To maximize the accuracy of predictions.

Q4. Which algorithm is commonly used for classification tasks insupervised learning?

(a) k-Means

(b) Decision Trees

(c) Principal Component Analysis (PCA)

(d) Hierarchical Clustering

Q5. What is the purpose of the training phase in supervised learning?

(a) To evaluate the model's performance.

(b) To make predictions on new, unseen data.

(c) To adjust the model's parameters based on labeled training data.

(d) To perform feature extraction.

Q6. Suppose you have the following dataset with two features (X1 and X2)and corresponding labels:

Data Point	X1	X2	Label
1	2.0	3.0	A
2	3.0	4.0	A
3	5.0	6.0	B
4	7.0	8.0	B
5	10.0	10.0	A

Now, consider a new data point with $X1$=6.0 and $X2$=7.0. Using kNN with k=3, predict the label for this new data point.

Q7. Suppose we have a dataset with two classes (A and B) and a node in a decision tree. In this node, there are 15 instances of class A and 10 instancesof class B. Calculate the Gini impurity for this node.

Q8. Suppose a decision tree is splitting a dataset based on a feature, and thedataset is initially impure with a Gini impurity of 0.6. After the split, the child nodes have the following impurities: Node 1 with 30 instances and Gini impurity of 0.2, Node 2 with 20 instances and Gini impurity of 0.3. Calculate the information gain.

Q9. In a decision tree, if each node is allowed to have a maximum depth of3, how many leaf nodes (terminal nodes) can the tree have at maximum?

Q10. Consider a dataset with a single feature (X) and corresponding targetvalues (Y):

Data Point	X	Y
1	2.0	5.0
2	4.0	8.0
3	6.0	12.0
4	8.0	15.0
5	10.0	20.0

Now, consider a new data point withX=7.0. Using kNN with

k=2, predict the target value (Y) for this new data point.

Ans:

1. B
2. D
3. C
4. B
5. C